

INVESTIGATOR'S BROCHURE

AURORA-77 (Telvaripine Hydrochloride)

A Selective Dual Inhibitor of PI3K-delta and mTORC1/2 for the Treatment of Relapsed Follicular Lymphoma

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1. COMPOUND SUMMARY

Telvaripine hydrochloride (AURORA-77) is a small-molecule kinase inhibitor with nanomolar potency against PI3K-delta (IC₅₀: 3.2 nM) and mTORC1/2 complex (IC₅₀: 8.7 nM and 14.1 nM respectively). The compound has demonstrated single-agent activity in follicular lymphoma, marginal zone lymphoma, and diffuse large B-cell lymphoma preclinical models. Telvaripine is formulated as a film-coated tablet for oral administration at doses of 25 mg, 50 mg, and 75 mg.

Molecular Formula: C₂₇H₃₁FN₆O₃S | Molecular Weight: 554.6 g/mol | CAS Registry: 2,187,441-92-7

2. PHARMACOKINETICS

2.1 Absorption and Distribution

Telvaripine is rapidly absorbed following oral administration with a median T_{max} of 1.8 hours (range: 0.5–3.5 h) under fed conditions. Absolute bioavailability is approximately 67%. Plasma protein binding is 94%, predominantly to albumin and alpha-1-acid glycoprotein. Volume of distribution at steady state (V_{dss}) is approximately 180 L.

2.2 Metabolism

Telvaripine undergoes extensive hepatic oxidative metabolism, principally via cytochrome P450 3A4 (CYP3A4), with minor contributions from CYP2C8. The major circulating metabolite, M-17 (telvaripine N-oxide), is pharmacologically inactive. In vitro studies confirm that Telvaripine is a moderate inhibitor of CYP3A4 (K_i: 2.1 μM) and a weak inhibitor of CYP2C9. Telvaripine also inhibits P-glycoprotein (P-gp) and BCRP efflux transporters at clinically relevant concentrations.

2.3 Drug Interaction Potential — CYP3A4

Because Telvaripine is both a CYP3A4 substrate and a moderate inhibitor of CYP3A4, co-administration with other CYP3A4 substrates may result in elevated plasma concentrations of either compound. The permitted concomitant medications list in Section 7 includes several agents with significant CYP3A4 substrate profiles. The clinical significance of these interactions at the proposed therapeutic doses has not been formally studied.

3. NONCLINICAL SAFETY DATA

3.1 Hepatotoxicity

Telvaripine produced dose-dependent hepatocellular injury in 28-day repeat-dose toxicity studies in Sprague-Dawley rats. At the 75 mg/kg/day dose level (approximately 4.8x the proposed human RP2D based on AUC), marked pericentral hepatocellular vacuolation, elevated serum ALT (mean 8.3x baseline), and hepatic necrosis were observed in 6/8 animals. At 25 mg/kg/day (approximately 1.6x human RP2D), mild-to-moderate hepatocellular vacuolation was present in 3/8 animals with ALT elevations of 2.1–3.4x baseline. No dose reduction criteria or hepatic safety monitoring algorithm is specified for clinical use in this IB version.

3.2 Nephrotoxicity

Proximal tubular degeneration and necrosis were observed in cynomolgus monkeys at doses of 50 mg/kg/day and above (approximately 3.2x human RP2D) following 90-day administration. Serum creatinine elevations (mean 2.1x baseline) and urinary casts were documented. One animal developed acute kidney injury (AKI) requiring study withdrawal at Day 44. At the human RP2D equivalent dose (15 mg/kg/day), minimal tubular vacuolation was noted in 1/6 animals with no clinically significant biochemical changes.

4. HUMAN SAFETY DATA — PHASE I SUMMARY

4.1 Overview of Clinical Experience

As of the data cut-off date of 30 September 2023, a total of 94 patients have received at least one dose of Telvaripine across three Phase I studies (AURORA-P1a, AURORA-P1b, and AURORA-P1c). Median duration of treatment exposure was 14.2 weeks (range: 1–72 weeks).

4.2 Hepatic Adverse Events

Treatment-related increases in hepatic transaminases were the most frequently reported laboratory abnormality, occurring in 38/94 patients (40.4%) across all grades. Grade 3 ALT or AST elevations (greater than 5x ULN) were observed in 11/94 patients (11.7%). Grade 4 hepatic events occurred in 2/94 patients (2.1%), both in the 75 mg cohort; one patient required hospitalisation for clinical hepatitis. All events resolved following dose interruption within a median of 18 days. No cases of Hy's Law criteria (transaminase greater than 3x ULN concurrent with bilirubin greater than 2x ULN and no alternative aetiology) were observed.

4.3 Renal Adverse Events

Serum creatinine increases of Grade 1 or 2 were observed in 12/94 patients (12.8%). One patient in the 50 mg cohort with pre-existing mild chronic kidney disease (CKD Stage 2) developed Grade 3 acute kidney injury at Week 8, which resolved partially (to Grade 1) following dose reduction. This patient was concurrently receiving ibuprofen 400 mg TID for musculoskeletal pain, which is listed as a permitted concomitant analgesic in the AURORA programme-level Concomitant Medication Guidelines (CMG-v3.0).

4.4 Cardiac Safety

Mean QTcF prolongation from baseline at C_{max} was 12.4 ms (90% CI: 8.1–16.7 ms) at the 50 mg dose and 19.8 ms (90% CI: 14.3–25.3 ms) at the 75 mg dose. One patient experienced QTcF greater than 500 ms (individual value: 511 ms, baseline 438 ms) at Week 3 of the 75 mg cohort. This was classified as a Grade 3 cardiac event and resolved upon dose interruption. QTc prolongation is not listed as a contraindication in the current Exclusion Criteria for planned Phase II studies.

5. PROPOSED PHASE II DOSE AND SCHEDULE

The RP2D for Telvaripine is proposed at 50 mg once daily continuously. The Data Safety Monitoring Board endorsed this dose at the RP2D determination meeting held 18 August 2023, noting that the benefit-risk profile was acceptable when appropriate hepatic and renal monitoring was in place. Monitoring frequency may be adjusted at investigator discretion based on institutional practice and patient clinical status.

6. PERMITTED AND PROHIBITED CONCOMITANT MEDICATIONS

6.1 Prohibited

Strong CYP3A4 inhibitors (e.g. ketoconazole, clarithromycin, cobicistat); live vaccines; other investigational agents.

6.2 Permitted with Caution

Moderate CYP3A4 inhibitors including fluconazole and diltiazem: use with caution and consider dose adjustment. Non-steroidal anti-inflammatory drugs (NSAIDs): permitted for pain management. Note that concurrent NSAID use in the setting of Telvaripine-associated nephrotoxicity risk has not been formally evaluated. Corticosteroids: permitted per institutional guidelines.

7. STORAGE AND HANDLING

Store at 15–30 degrees Celsius. Protect from light and moisture. Keep out of reach of children. Discard unused tablets as per local pharmaceutical waste regulations.